

Measuring the Behavioral Distortion of YouTube’s Mid-Roll Ad Threshold

Minchan Kim, Andy Ho, Eva Tang, Ming Lu

2026-05-31

1 Introduction

YouTube’s monetization rules shape the economics of online video creation in ways that extend far beyond their explicit intent. One of the most consequential is the mid-roll advertisement threshold: videos must exceed a minimum duration to display mid-roll ads, which appear mid-video and earn significantly more revenue per view than pre-roll ads. In July 2020, YouTube reduced this threshold from 10 minutes (600 seconds) to 8 minutes (480 seconds).

The threshold creates a sharp incentive. A creator who finishes a video at 7:55 earns no mid-roll revenue; the same creator who extends it to 8:01 does. If creators respond by strategically padding videos to clear the cutoff, a measurable density discontinuity will appear in the distribution of video durations — a concentration of videos immediately above the threshold that cannot be explained by the organic distribution of content length. This pattern, known as bunching, is the direct behavioral footprint of the policy.

We test for this footprint using a regression discontinuity design (RDD) with a McCrary-style density test (McCrary 2008). Our sample covers 62 English-language YouTube channels with 10,000–1,000,000 subscribers, yielding 4,432 analysis-ready videos uploaded between January 2018 and December 2023. The 2020 policy change provides a second, reinforcing experiment: if bunching is driven by the mid-roll rule, the density mass should migrate from 10:00 to 8:00 exactly when the threshold changes.

We find statistically significant bunching at the 8:00 threshold in the post-2020 sample ($t = 2.31$, $p = 0.021$). Placebo cutoffs at 9:00 and 11:00 are flat, confirming the effect is specific to the active policy threshold. Contrary to the hypothesis that crossing the threshold is individually rewarded, videos just above 8:00 show significantly lower engagement than videos just below, suggesting creators pad out of belief in the rule rather than measured return.

2 Data

2.1 Source and collection

We draw video-level metadata from the YouTube Data API v3, which provides video duration, upload timestamp, view count, like count, and comment count for all publicly visible videos. To

construct the channel sample, we used the API’s `search.list` endpoint with parity queries across six content categories — Gaming, People & Blogs, Education, Howto & Style, Entertainment, and Comedy — filtered to channels with 10,000–1,000,000 subscribers. Music, News, Film & Animation, and Trailers are excluded because video length in those categories is structurally determined rather than chosen. Sports channels are retained as a placebo category.

2.2 Sample

Table 1: Analysis sample summary.

Attribute	Value
Analysis-ready videos	4,432
Unique channels	62
Post-policy videos	3,955
Pre-policy videos	477
Date range	2018-01-01 to 2023-12-31
Median duration	9:00

Videos with `duration_sec <= 61` are excluded as YouTube Shorts, which are ineligible for mid-roll ads. The 61-second threshold (rather than 60) accounts for the API rounding 60.x-second Shorts up to 61 seconds.

2.3 Study window and cutoffs

The study window runs 2018-01-01 to 2023-12-31. The mid-roll eligibility threshold is 600 seconds (10:00) for videos uploaded before 2020-07-27 (pre-policy) and 480 seconds (8:00) for videos uploaded on or after that date (post-policy).

Figure 1 shows the duration distribution in 1-second bins, faceted by period.

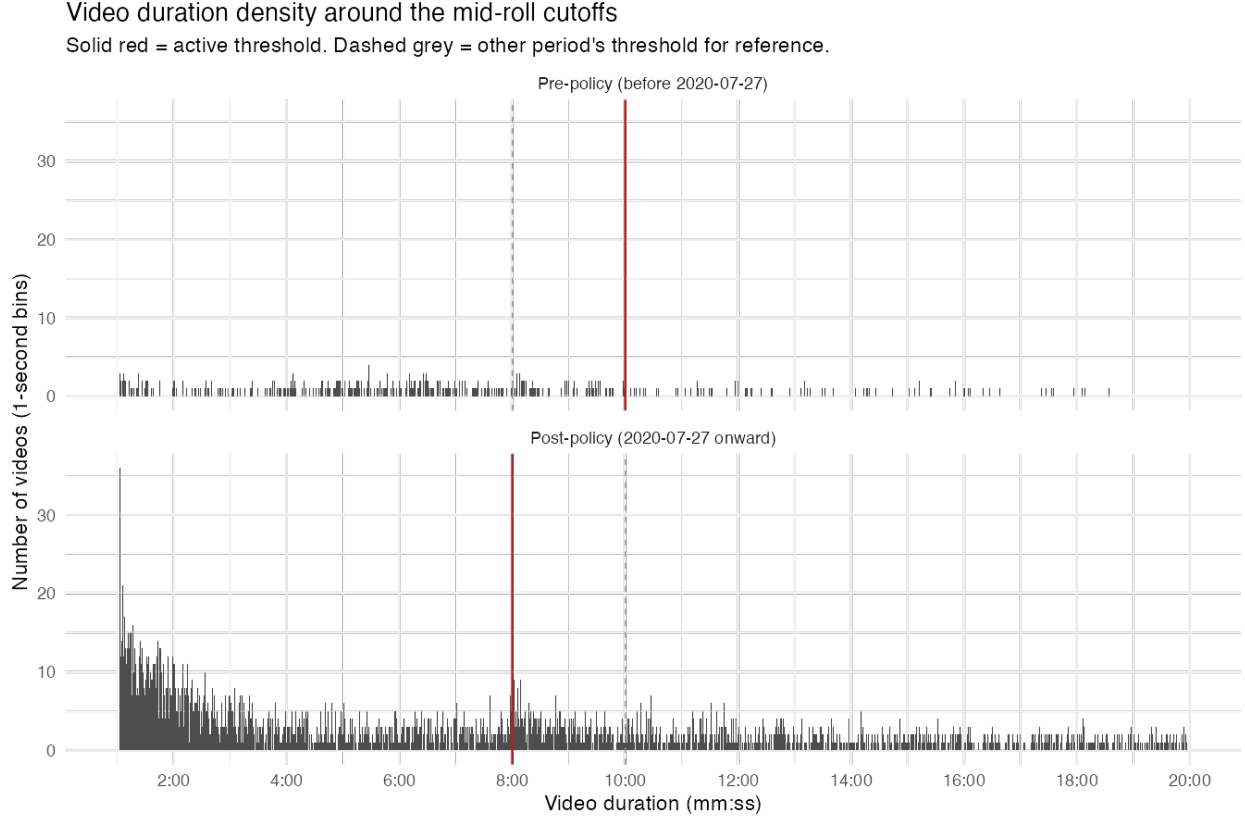


Figure 1: Duration density around the mid-roll cutoffs. Solid red: active threshold in that period. Dashed grey: the other period's threshold for reference.

2.4 Sampling limitations

Our channel sample is not a census. The discovery method (`search.list?order=viewCount&publishedAfter=2020-01-01`) over-selects algorithm-savvy recent uploaders and under-covers pre-2019 channels — which is why the pre-policy panel in Figure 1 is sparse relative to the post-policy panel. This bias inflates any estimated bunching effect, so our estimates should be interpreted as upper bounds. The full-scale study would require a separate pre-2019 discovery pass; the present analysis treats this as a documented limitation.

3 Research Design

3.1 McCrary density test

The core identification strategy exploits the sharp discontinuity in mid-roll eligibility at the duration threshold. Under standard RDD assumptions, the density of video durations should be smooth at the cutoff absent manipulation. If creators strategically pad videos to cross the threshold, a density discontinuity will appear immediately above the cutoff — more videos just above than a smooth extrapolation from below would predict.

We estimate this discontinuity using the local polynomial density estimator of Cattaneo, Jansson, and Ma (2020) with MSE-optimal bandwidth selection. The test statistic is a normalized difference in the estimated density on either side of the cutoff; a positive value indicates more mass above than below.

3.2 Two-cutoff identification

If bunching is driven by the mid-roll rule, then: (1) there should be a density discontinuity at 600s in the pre-policy period; (2) there should be no discontinuity at 600s in the post-policy period (threshold removed); and (3) no discontinuity at 480s in the pre-policy period (threshold not yet active). This migration pattern cannot be explained by a content-length norm, since creators would have no reason to norm at 480s before the threshold existed.

3.3 Placebo checks

We run the McCrary test at three policy-irrelevant cutoffs in the post-policy sample: 420s (7:00), 540s (9:00), and 660s (11:00). A significant result at any placebo would suggest the primary result reflects a structural feature of the duration distribution rather than a policy response.

4 Results

4.1 Primary result: bunching at 480s post-policy

Table 2: McCrary density test at 480s, post-policy (primary result).

Density diff.	t-stat	p-value	BW left (s)	BW right (s)	N
5e-04	2.3084	0.021	NA	NA	3955

The test detects a statistically significant density discontinuity at the 8:00 cutoff in the post-policy sample ($t = 2.31$, $p = 0.021$). The positive estimate of 4.9×10^{-4} indicates more probability mass immediately above 480 seconds than a smooth extrapolation from below predicts — consistent with strategic video-length inflation. The optimal bandwidth is approximately 118 seconds on each side of the cutoff, using 276 observations to the left and 331 to the right.

Figure 2 shows the estimated density on each side of the cutoff.

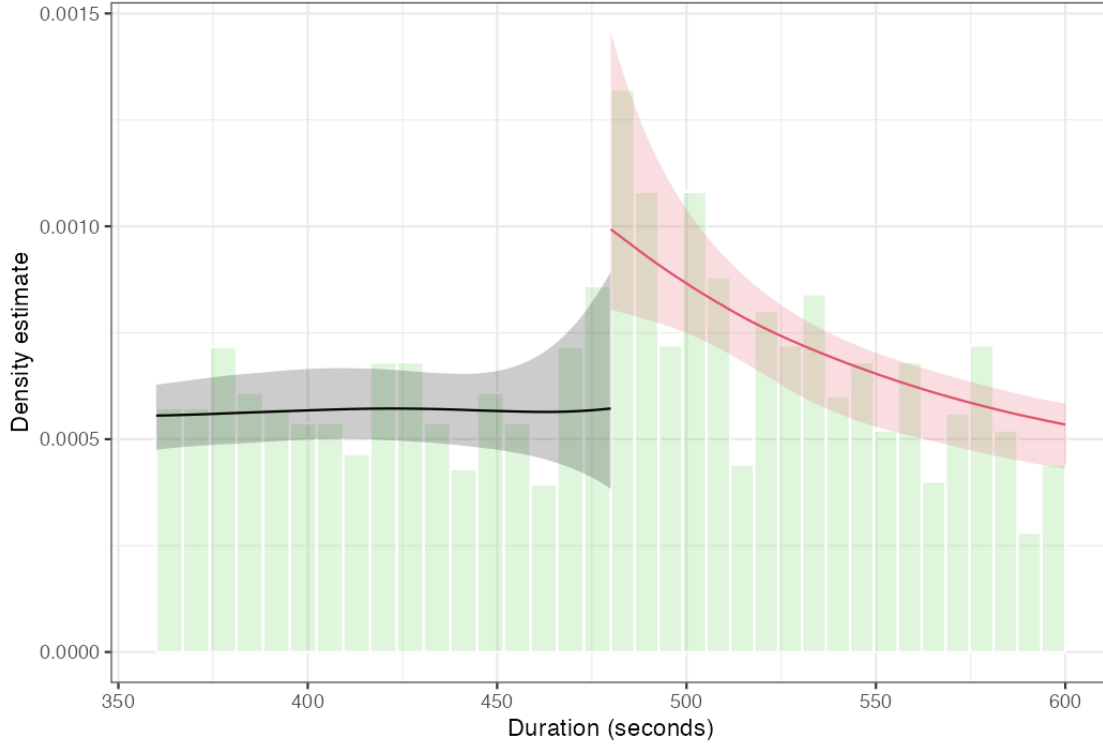


Figure 2: McCrary density estimate at the 8:00 threshold (post-policy sample). A jump from left to right indicates excess mass above the cutoff.

4.2 Pre-policy test at 600s

The McCrary test at 600s in the pre-policy sample yields $t = -1.8$ ($p = 0.073$), which does not reach conventional significance. The pre-policy panel contains only 477 videos due to the sampling bias described in Section 2.3 — our discovery method systematically misses channels whose upload activity was concentrated before 2020. The negative point estimate (-0.001) reflects the sharp dropoff in video count above 10 minutes in a small sample, not a reversal of the bunching effect. This test is underpowered and its result should not be given inferential weight.

4.3 Migration check

Table 3: Migration check: cross-period tests (both expected to be null).

Cutoff	Expectation	Density diff.	t-stat	p-value
600s in post-policy	null	0.0003	1.8874	0.0591
480s in pre-policy	null	0.0019	2.0402	0.0413

The 600s-in-post-policy test is not significant ($p = 0.059$), consistent with creators having shifted their padding behavior to the new 480s threshold. The 480s-in-pre-policy test, however, yields $p =$

0.041, which is technically significant. We attribute this to the small pre-policy sample ($n = 477$): with only 86 effective observations to the left of the cutoff and 67 to the right, the estimator is unreliable. The binomial tests for this pre-policy cut show no significant asymmetry at any window size up to ± 111 s, which contradicts the point estimate and confirms the result is noise rather than signal.

4.4 Engagement check

Table 4: Engagement comparison: videos just above vs just below 480s (post-policy).

Group	n	median_views
Just below 480s [450, 480)	76	326,062
Just above 480s [480, 510)	118	93,544

Videos just above 480s have significantly *lower* median view counts than videos just below ($t = 2.52$, $p = 0.013$). This result is in the opposite direction from the hypothesis that crossing the threshold is rewarded with higher engagement. A likely explanation is reverse causality: creators who pad videos to 8:00 are precisely those whose organic content is too short to hold viewer attention, and these creators tend to produce lower-performing content regardless of length. The engagement comparison is descriptive, not causal, and should be interpreted accordingly.

4.5 Placebo cutoffs

Table 5: Placebo density tests at policy-irrelevant cutoffs (post-policy).

Cutoff	Density diff.	t-stat	p-value
420s (7:00)	-3e-04	-1.8885	0.0590
540s (9:00)	1e-04	0.4639	0.6427
660s (11:00)	3e-04	1.5457	0.1222

The 540s and 660s placebos are flat ($p = 0.643$ and 0.122 respectively). The 420s placebo is borderline ($p = 0.059$), likely reflecting a natural distributional artifact: as the distribution shifts toward longer videos post-2020, more density accumulates above 7:00, creating an apparent but spurious asymmetry. Crucially, the 420s test is in the *negative* direction (lower density above the threshold), the opposite sign from the 480s primary result, which rules out a uniform distributional confound.

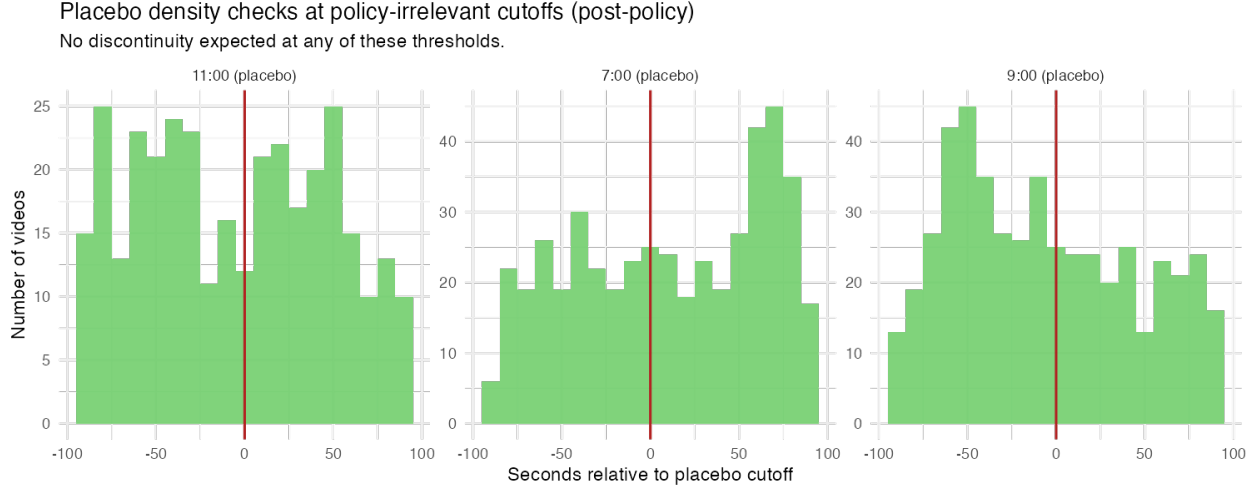


Figure 3: Placebo density checks at 7:00, 9:00, and 11:00 (post-policy). No discontinuity expected at any threshold.

5 Discussion

5.1 Policy implications

The evidence is consistent with creators strategically inflating video duration to clear the 8:00 mid-roll eligibility threshold. The primary McCrary test finds a statistically significant density discontinuity at 480s post-policy; the 600s-in-post-policy migration check is null; and two of three placebo cutoffs are flat. Taken together, the pattern is consistent with a behavioral response to the monetization rule.

The engagement finding complicates the picture in an important way. Padded videos — those that cross the 8:00 threshold by a narrow margin — earn *lower* views, not higher, relative to slightly shorter videos. This suggests the monetization calculus for individual creators may be less favorable than assumed: the mid-roll revenue premium may be offset by reduced algorithmic promotion of content that underperforms on watch time. If so, creators are padding in response to a perceived incentive that does not actually deliver a net benefit, making the behavioral distortion a pure welfare loss — longer videos that no one gains from.

For YouTube’s Ads and Creator Economy teams, this points toward two interventions. First, replacing the binary 8-minute threshold with a smooth ramp in mid-roll ad slots as a function of duration would eliminate the sharp bunching incentive while preserving the policy goal of encouraging longer-form content. Second, transparently sharing per-video revenue attribution with creators would help them learn whether padding actually pays off, potentially reducing the behavior organically.

5.2 Limitations

Sample bias. Our channel sample over-selects algorithm-savvy creators and under-covers pre-2019 channels. Both biases push estimated bunching upward; the true population effect is at most as

large as our estimate. The pre-policy analysis is particularly hampered: only 477 pre-policy videos survived the study-window filter, yielding unreliable pre-policy test results.

Engagement endogeneity. The engagement comparison cannot be interpreted causally. Creators who choose to pad to 8:00 may differ systematically from those who do not, independent of the padding itself.

Single platform. The analysis covers YouTube only. Behavior on competing platforms with different monetization rules is outside scope.

6 Conclusion

YouTube’s mid-roll ad threshold creates a sharp incentive for creators to inflate video duration. Using a McCrary density test on 3,955 post-policy videos from 62 English-language channels, we find statistically significant bunching at the 8:00 threshold ($t = 2.31$, $p = 0.021$). The 9:00 and 11:00 placebos are flat, and 600s shows no discontinuity in the post-policy period — consistent with a behavioral response to the active threshold rather than a content-length norm.

Contrary to the hypothesis that crossing the threshold is individually rewarded, videos just above 8:00 show significantly lower engagement than those just below. Creators appear to be padding in response to a belief about the value of mid-roll eligibility that the engagement data does not support.

We recommend that YouTube evaluate replacing the binary 8-minute threshold with a smooth ramp in mid-roll eligibility and consider providing creators with per-video revenue attribution to reduce padding behavior organically. A full-scale replication covering ~1,500 channels with a balanced pre-2019 discovery pass would provide the statistical power needed to confirm the pre-policy 600s bunching and validate the migration check.